GRIFFIN Data Format Specification

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The GRIFFIN data acquisition system runs under the MIDAS Data Acquisition system in order to record the data from the GRIFFIN HPGe spectrometer for decay spectroscopy experiments at TRIUMF-ISAC.

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Chapter 1

Introduction

1.1 Binary Counting Definition

The bits in the words in this document are counted from 0 to 7 in each byte. Therefore a 4-byte, 32 bit word contains bits from 0 to 31.

An example of how the bits in a 16-bit, 2-byte word are counted in this document:

Bit Number:	15	14	13	12	11	10	09	08	07	06	05	04	03	02	01	00
Bit Weight:	32,768				2,048				128				8			

Chapter 2

GRIF-16

2.1 Event Format and Size

Detectors counting at a high rate will register multiple hits during a fixed evaluation period. In order to take advantage of higher counting rates, it is therefore essential to handle these pile-up events and recover the energies and times for every interaction in the detector. A method for extracting these energies and times is shown in figure 2.1 which involves splitting the integration window of size, K, into multiple windows, K1, K2 etc. The event can also be flagged and the waveform output for evaluation offline.

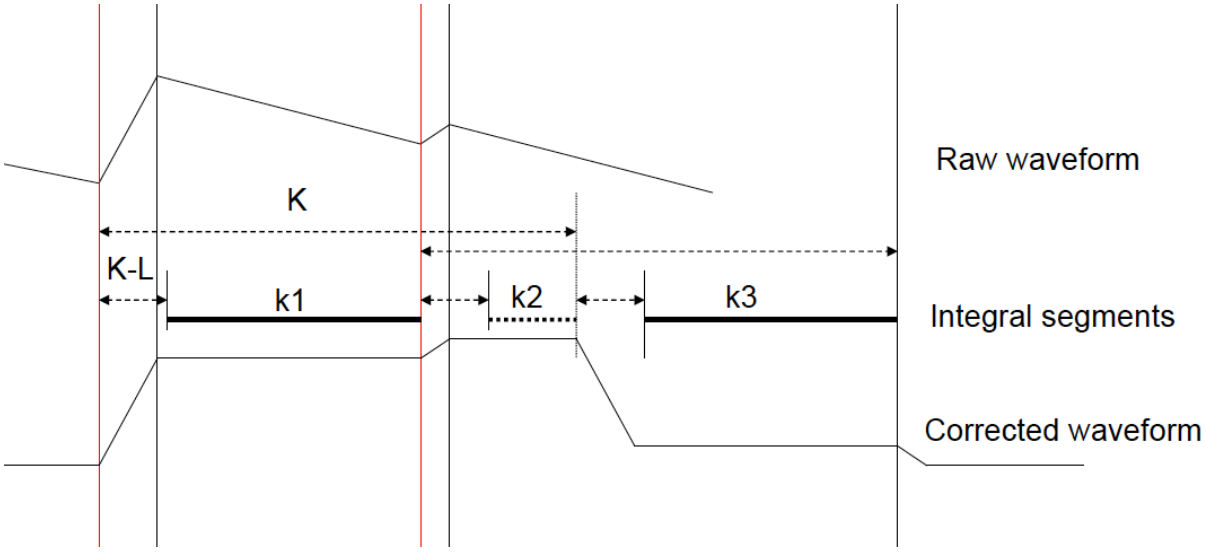


Figure 2.1: Method for extracting energy and time for pile-up events in the firmware.

The event format for handling pileup events is of variable size and depends on the number of hits in the evaluation window, the maximum is 3 hits in one window. The format is based on 4-byte (32-bit) words shown in the following tables, the number of bits for each sub-word is indicated in brackets. Waveforms can also be included at 2 bytes per sample.

Table 2.1: GRIFFIN GRIF-16 Event Format written to disk, bank GRF2:

I	Packet Type (4) **0x8**	Pile-Up Type (2)	Data Type (3)	Num. Filter Patterns (3)	Address (16)	Det. Type (4)	4 bytes
II	Fixed 00 (2)	Master Filter Patterns Passed (14)		PPG Pattern (16)			4 bytes
III	Fixed 0 (1)	Master Filter ID (31)					4 bytes
IV	Packet Type (4) **0x9**	Channel Trigger ID (28)					4 bytes
V	Packet Type (4) **0xa**	Time Stamp low bits (28)					4 bytes
VI	Packet Type (4) **0xb**	Deadtime (14)		Time Stamp high bits (14)			4 bytes
VIa	Packet Type (4) **0xd**	Network Packet ID (28)					4 bytes
VIb	Packet Type (4) **0xc**	Waveform Sample (28)					4 bytes
VII	Fixed 0 (1)	Upper K1 bits (5)	K1 Pulse Height (26)				4 bytes
VIII	Fixed 0 (1)	Lower K1 bits (5)	K1 Amplitude-Corrected Timing (26)				4 bytes
IX	Fixed 0 (1)	Upper K2 bits (5)	K2 Pulse Height (26)				4 bytes
X	Fixed 0 (1)	Lower K2 bits (5)	K2 Amplitude-Corrected Timing (26)				4 bytes
XI	Fixed 0 (1)	Upper K3 bits (5)	K3 Pulse Height (26)				4 bytes
XII	Fixed 0 (1)	Lower K3 bits (5)	K3 Amplitude-Corrected Timing (26)				4 bytes
XIII	Packet Type (4) **0xe**	Event Trailer (10)		Event Trailer - Repeated Time Stamp low bits (14)			4 bytes

Total one hit	9 words I-VIII+XIII:	36 bytes
Total two hits	11 words I-X+XIII:	44 bytes
Total three hits	13 words I-XIII:	52 bytes

2.2 Bank version information

Over the lifetime of the data acquisition system, the data format is expected to change. Care will be taken to make these changes backward compatible, but occasionally, this will not be possible. In that case, the bank name will be used to track the data format used. This document will be maintained with the current data format. Differences between the current data format and the older data formats will be detailed in this section. The current bank name is GRF2.

- GRF2: current bank name
- GRF1: different header format than GRF2. Header word contains the following information (bit size): Packet Type (4), Number Filter Patterns (4), Data Type (3), Pile-Up Type (3), Address (14), Detector Type (4).

2.3 Pile-Up Type

The Pile-Up Type indicates how many hits were recorded within the evaluation window, with the maximum being 3 hits in one window. The Pile-Up Type is provided in word I as bits 18-20 (2 bits).
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF0000000	>>	28	=	0x8	=	8

Example of extracting the Pile-Up Type:

Word I		Mask		Shift		Hex result		Decimal result
0x80007C80	&	0x0C000000	>>	26	=	0x0	=	0 (1 Hit)
0x84007C80	&	0x0C000000	>>	26	=	0x1	=	1 (2 Hit)
0x88007C80	&	0x0C000000	>>	26	=	0x2	=	2 (3 Hits)
0x8C007C80	&	0x0C000000	>>	26	=	0x3	=	3 (>3 Hits = Bad)

2.4 Data Type

The Data Type indicates where the data was generated (GRIF-16, GRIF-4G, GRIF-C etc.) and is provided in word I as bits 23-25 (3 bits). The packet type is identified in the 4 upper bits of word I as type 8.
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF0000000	>>	28	=	0x8	=	8

The Data Type number can be extracted as:

Word I		Mask		Shift		Hex result		Decimal result
0x80A4A998	&	0x03800000	>>	23	=	0x1	=	1 (GRIF-16)
0x8124A998	&	0x03800000	>>	23	=	0x2	=	2 (GRIF-4G)
0x81A4A998	&	0x03800000	>>	23	=	0x3	=	3 (GRIF-C Slave)
0x8234A998	&	0x03800000	>>	23	=	0x4	=	4 (GRIF-C Master)

2.5 Number of Filter Patterns

The Number of Filter Patterns which have been satisfied by this event fragment is provided in word I as bits 20-22 (3 bits). The packet type is identified in the 4 upper bits of word I as type 8.
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF0000000	>>	28	=	0x8	=	8

The number of filter patterns can be extracted as:

Word I		Mask		Shift		Hex result		Decimal result
0x8614A998	&	0x00700000	>>	20	=	0x1	=	1 (1 pattern word)
0x8624A998	&	0x00700000	>>	20	=	0x2	=	2 (2 pattern words)
0x8634A998	&	0x00700000	>>	20	=	0x3	=	3 (3 pattern words)

2.6 Address

The Address is provided in word I as bits 4 to 19 (16 bits). This indicates which channel on which module has provided this event fragment.
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF0000000	>>	28	=	0x8	=	8

An example of unpacking the Address from the header word:

Word I		Mask		Shift		Hex result		Decimal result
0x82228280	&	0x000FFFF0	>>	4	=	0x0505	=	1285

The various properties with the Address word are as follows:

Property	Size	Values	Description
M	4 bits	0 to 15	Input channel to Master or ID number of Slave
S	4 bits	0 to 15	Input channel to Slave or ID number of Digitizer
C	8 bits	0 to 255	Input channel to Digitizer

These properties can be extracted from this 16 bit word in the following way:

Address		Mask		Shift		Hex result		Decimal result
0x1405	&	0x0000F000	>>	12	=	0x01	=	1 (M)
0x1405	&	0x00000F00	>>	8	=	0x04	=	4 (S)
0x1405	&	0x000000FF	>>	0	=	0x05	=	5 (C)

2.7 Detector Type

The Detector Type is provided in word I as bits 0 to 3 (4 bits). This indicates which detector type has generated this event fragment and is used primarily in master filter decisions.
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A928	&	0xF0000000	>>	28	=	0x8	=	8

The Detector Type value can be extracted as:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A900	&	0x0000000F	>>	0	=	0x00	=	00 (HPGe Low Gain)
0x8074A902	&	0x0000000F	>>	0	=	0x02	=	02 (SCEPTAR)
0x8074A905	&	0x0000000F	>>	0	=	0x05	=	05 (Si(Li))

Here is a full list of GRIFFIN Detector Types:

ID	**Short Name**	**Full Name**	**Description**
00	GRGa	Ge	16 HPGe Clovers Low Gain
01	GRGb	Ge	16 HPGe Clovers High Gain
02	SEP	Beta	20 plastic scintillators (SCEPTAR)
03	DAN	DANTE	8 LaBr3 scintillators Energy signals
04	DAT	DANTE	8 LaBr3 scintillators TAC signals
05	PAC	PACES	5 SiLi for ICE
06	DSC	DESCANT	70 Scintillators for neutrons
07	GRS	Suppressors	16 sets of BGO for clovers
08	DAS	Suppressors	8 sets of BGO for DANTE
09	SET	Beta	Zero-Degree scintillator (ZDS)
10	DSC	DESCANT (Gamma)	Gamma identified by pulse shape
11	DSC	DESCANT (Neutron)	Neutron identified by pulse shape
12			Undefined
13			Undefined
14			Undefined
15			Undefined

2.8 Master Filter ID

The Master Filter ID is provided in word III as 31 bits. The Master Filter ID is an event counter which is reset at the start of a run and increments by 1 each time a new event master filter condition is passed in the master filter algorithm in the

Master GRIF-C. The maximum value is 2147483649 above which the counter will wrap around.
Example of extracting the Master Filter ID:

Word II		Mask		Shift		Hex result		Decimal result
0x56007C84	&	0x7FFFFFFF	>>	0	=	0x56007C84	=	1442872452

2.9 Master Filter Pattern

There are up to eight (more in the future) Filter Patterns which can be set and tested for every fragment. The Identifier(s) (1-8) of the Filter Pattern which was passed is provided in word II as bits 16 to 30 (14 bits). If multiple Filter conditions were passed by this fragment then there will be multiple Filter Patterns indicated in this word.
Example of extracting the Master Filter Pattern(s):

Word III		Mask		Shift		Hex result		Decimal result
0x00010F81	&	0x3FFF0000	>>	16	=	0x0001	=	1 (Only Filter 1 passed)
0x00050F81	&	0x3FFF0000	>>	16	=	0x0005	=	5 (Filters 1 and 3 passed)
0x00850F81	&	0x3FFF0000	>>	16	=	0x0085	=	133 (Filters 1, 3 and 8 passed)

2.10 PPG Pattern

The PPG Pattern is a recording of the Programable Pattern Generator status at the time that the threshold crossing occurred, and is provided in word II as bits 0 to 15 (16 bits). The pattern represents, for example, if the beam were being delivered or if it was blocked for decay counting etc.
Example of extracting the PPG Pattern:

Word III		Mask		Shift		Hex result		Decimal result
0x0F810F81	&	0x0000FFFF	>>	0	=	0x00000F81	=	3969

2.11 Channel Trigger ID

The Channel Trigger ID is provided in word IV as the 28 lower bits. The Channel Trigger ID is an event counter which is reset at the start of a run and increments by 1 each time a new threshold crossing occurs for this channel on the GRIF-16 or GRIF-4G. The word type is identified in the 4 upper bits as type 9.
Example of identifying the Channel Trigger ID word type:

Word IV		Mask		Shift		Hex result		Decimal result
0x900461FF	&	0xF0000000	>>	28	=	0x9	=	9

The Channel Trigger ID number can wrap around if it becomes too large. The user should monitor the channel trigger id numbers which should increase throughout the run. The maximum value of channel trigger id is 268435455. If the number changes from very large to small then the counter has wrapped around, this should be tested at every event.
Example of unpacking the Channel Trigger ID:

Word IV		Mask		Shift		Hex result		Decimal result
0x900461FF	&	0x0FFFFFFF	>>	0	=	0x000461FF	=	287231

2.12 Timestamp

The timestamp is stored in two separate words as the lower 28 bits and 14 bits of the two words, word V and word VI respectively. Word V is the lower bits of the Timestamp and Word VI is the upper bits of the Timestamp. The type is identified in the 4 upper bits of both words as type 10 for the lower bits and 11 for the upper bits.
Example of identifying a Timestamp word type:

Word V/VI		Mask		Shift		Hex result		Decimal result
0xA074A998	&	0xF0000000	>>	28	=	0xA	=	10 (lower bits)
0xB074A998	&	0xF0000000	>>	28	=	0xB	=	11 (upper bits)

An example of unpacking the Timestamp value:

Word V/VI		Mask		Shift		Hex result		Decimal result
0xA074A998	&	0x0FFFFFFF	>>	0	=	0x0074A998	=	7645592
0xB0000005	&	0x00003FFF	>>	0	=	0x00000005	=	5

The upper and lower bits of the Timestamp can then be combined to make the full 42-bit timestamp. This is done in two steps by shifting the upper bits and then adding the lower bits using a bitwise or (||):

Build Timestamp				Hex result		Decimal result
0x00000005	<<	28	=	0x05000000		
0x05000000	||	0x0074A998	=	0x0574A998	=	91531672

The timestamp is in units of 10 nanoseconds and has a maximum value of 4,398,046,511,103. The timestamp will therefore reset either after 509 days or when the GRIF-C Master module is power cycled.

The 28 lower bits are also stored in the event trailer.

2.13 Deadtime

There are two methods of determining the event-by-event deadtime of the system for precision measurements; firstly to operate at a fixed deadtime for every event, or

secondly to measure and record the deadtime of each event. Both methods will be trialled and evaluated as part of the GRIFFIN DAQ development.

The deadtime for each event is provided as bits 14-27 (14 bits) of word VII. The type is identified in the 4 upper bits as type 0xB.
Example of identifying a Deadtime word type:

Word VII		Mask		Shift		Hex result		Decimal result
0xB074A998	&	0xF0000000	>>	28	=	0xB	=	11

An example of unpacking the Deadtime value:

Word VII		Mask		Shift		Hex result		Decimal result
0xC074A998	&	0x0FFFC000	>>	14	=	0x001D2	=	466

The deadtime is in units of 10 ns and the maximum value is 16,383 representing 162.383 microseconds of deadtime for this event readout.

2.14 Network Packet ID

The Network Packet ID is currently provided in word VIa as the 28 lower bits. The Network Packet ID is an event counter which is reset at the start of a run and increments by 1 each time a new event is dispatched by the GRIF-C master. The word type is identified in the 4 upper bits as type 0xD.
Example of identifying the Network Packet ID word type:

Word VIa		Mask		Shift		Hex result		Decimal result
0xD00461FF	&	0xF0000000	>>	28	=	0xD	=	13

The Network Packet ID number can wrap around if it becomes too large. The user should monitor the Network Packet id numbers which should increase throughout the run. The maximum value of Network Packet ID is 268435455. If the number changes from very large to small then the counter has wrapped around, this should be tested at every event.
Example of unpacking the Network Packet ID:

Word VIa		Mask		Shift		Hex result		Decimal result
0xD00461FF	&	0x0FFFFFFF	>>	0	=	0x000461FF	=	287231

2.15 Waveform Data

The Waveform Data is provided in multiple words, the number of which depends on the waveform length. These words are inserted into the event between words VI and VII. Waveform data words can be identified by their type which is in the 4 upper bits of each words as type C.

Word		Mask		Shift		Hex result		Decimal result
0xC57715CC	&	0xF0000000	>>	28	=	0xC	=	12

The Waveform Data values are in units of ADC bins.

2.16 K Values

The 10 bits of the K value are provided in words VII and VIII as bits 26-30.
Example of extracting the K value:

Word VII/VIII		Mask		Shift		Hex result		Decimal result
0x2074A998	&	0x7C000000	>>	26	=	0x08	=	8
0x5874A998	&	0x7C000000	>>	26	=	0x16	=	22

The upper and lower bits of the K value can then be combined to make the full K value. This is done in two steps by shifting the upper bits and adding the lower bits:

Build K Value				Hex result		Decimal result
0x08	<<	5	=	0x100		
0x100	||	0x16	=	0x116	=	278

2.17 Pulse Height Signal

The pulse height signal is provided in word VII as the 26 lower bits, which represent Pulse height 1. The maximum value is 67108863 and is in units of ADC channels.

Two examples of unpacking the Pulse Height value:

Word VII		Mask		Shift		Hex result		Decimal result
0x680001FF	&	0x03FFFFFF	>>	0	=	0x000001FF	=	511
0x680005B5	&	0x03FFFFFF	>>	0	=	0x000005B5	=	1461

For pile-up events the charge value of the secondary interactions are given in words X and XII.

2.18 Amplitude-Corrected Timing Signal

The timing from a Amplitude-Corrected Timing algorithm is provided in word VIII as the 26 lower bits. The maximum value is 67108863 and is in units of 10 nanoseconds.
Two examples of unpacking the Amplitude-Corrected timing value:

Word VIII		Mask		Shift		Hex result		Decimal result
0x78000052	&	0x03FFFFFF	>>	0	=	0x00000052	=	82
0x780000F6	&	0x03FFFFFF	>>	0	=	0x000000F6	=	246

For pile-up events the Amplitude-Corrected timing values of the secondary interactions are given in words VIII and XII.

2.19 Event Trailer

The Event Trailer is provided in word XIII and it indicates the end of the event fragment. The type is identified in the 4 upper bits as type 14.
Example of identifying the Event Trailer word type:

Word XIII		Mask		Shift		Hex result		Decimal result
0xEEEEEEEE	&	0xF0000000	>>	28	=	0xE	=	14

The 14 lower bits of the timestamp is repeated in the event trailer. This can be used as an additional check to ensure the event is not corrupted. An example of unpacking the lower bits of the Timestamp value from the Event Trailer word:

Word XIII		Mask		Shift		Hex result		Decimal result
0xE074A998	&	0x0003FFFF	>>	0	=	0x0000A998	=	43416

Chapter 3

GRIF-4G

The format from this digitizer will be similar to that of the GRIF-16 but additional information may be provided associated with DESCANT detectors. This information will include output from additional signal-processing modules such as a neutron-gamma discrimination algorithm.

Table 3.1: GRIFFIN GRIF-4G Event Format written to disk, bank GRF2:

I	Packet Type (4) **0x8**	Pile-Up Type (2)	Data Type (3)	Num. Filter Patterns (3)	Address (16)	Det. Type (4)	4 bytes
II	Fixed 00 (2)	Master Filter Patterns Passed (14)		PPG Pattern (16)			4 bytes
III	Fixed 0 (1)	Master Filter ID (31)					4 bytes
IV	Packet Type (4) **0x9**	Channel Trigger ID (28)					4 bytes
V	Packet Type (4) **0xa**	Time Stamp low bits (28)					4 bytes
VI	Packet Type (4) **0xb**	Deadtime (14)		Time Stamp high bits (14)			4 bytes
VIa	Packet Type (4) **0xd**	Network Packet ID (28)					4 bytes
VIb	Packet Type (4) **0xc**	Waveform Sample (28)					4 bytes
VII	Fixed 0 (1)	Upper K1 bits (5)	K1 Pulse Height (26)				4 bytes
VIII	Fixed 0 (1)	Lower K1 bits (5)	K1 CFD Timing (26)				4 bytes
IX	Packet Type (4) **0xf**	Upper Long Integration bits (9)	Short Charge Integration (19)				4 bytes
X	Fixed 0 (1)	Lower Long Integration bits (10)	Zero-Crossing Time (21)				4 bytes
XI	Packet Type (4) **0xe**	Event Trailer (10)		Event Trailer - Repeated Time Stamp low bits (14)			4 bytes

Total one hit 11 words I-VIII, XI: 44 bytes

Total one hit (DESCANT Detector Type only) 13 words I-XI: 52 bytes

3.1 Bank version information

Over the lifetime of the data acquisition system, the data format is expected to change. Care will be taken to make these changes backward compatible, but occasionally, this will not be possible. In that case, the bank name will be used to track the data format used. This document will be maintained with the current data format. Differences between the current data format and the older data formats will be detailed in this section. The current bank name is GRF2.

- GRF2: current bank name
- GRF1: different header format than GRF2. Header word contains the following information (bit size): Packet Type (4), Number Filter Patterns (4), Data Type (3), Pile-Up Type (3), Address (14), Detector Type (4).

3.2 Pile-Up Type

The Pile-Up Type indicates how many hits were recorded within the evaluation window, with the maximum being 3 hits in one window. The Pile-Up Type is provided in word I as bits 18-20 (2 bits).
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF0000000	>>	28	=	0x8	=	8

Example of extracting the Pile-Up Type:

Word I		Mask		Shift		Hex result		Decimal result
0x80007C80	&	0x0C000000	>>	26	=	0x0	=	0 (1 Hit)
0x84007C80	&	0x0C000000	>>	26	=	0x1	=	1 (2 Hit)
0x88007C80	&	0x0C000000	>>	26	=	0x2	=	2 (3 Hits)
0x8C007C80	&	0x0C000000	>>	26	=	0x3	=	3 (>3 Hits = Bad)

3.3 Data Type

The Data Type indicates where the data was generated (GRIF-16, GRIF-4G, GRIF-C etc.) and is provided in word I as bits 23-25 (3 bits). The packet type is identified in the 4 upper bits of word I as type 8.
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF0000000	>>	28	=	0x8	=	8

The Data Type number can be extracted as:

Word I		Mask		Shift		Hex result		Decimal result
0x80A4A998	&	0x03800000	>>	23	=	0x1	=	1 (GRIF-16)
0x8124A998	&	0x03800000	>>	23	=	0x2	=	2 (GRIF-4G)
0x81A4A998	&	0x03800000	>>	23	=	0x3	=	3 (GRIF-C Slave)
0x8234A998	&	0x03800000	>>	23	=	0x4	=	4 (GRIF-C Master)

3.4 Number of Filter Patterns

The Number of Filter Patterns which have been satisfied by this event fragment is provided in word I as bits 20-22 (3 bits). The packet type is identified in the 4 upper bits of word I as type 8.
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF0000000	>>	28	=	0x8	=	8

The number of filter patterns can be extracted as:

Word I		Mask		Shift		Hex result		Decimal result
0x8614A998	&	0x00700000	>>	20	=	0x1	=	1 (1 pattern word)
0x8624A998	&	0x00700000	>>	20	=	0x2	=	2 (2 pattern words)
0x8634A998	&	0x00700000	>>	20	=	0x3	=	3 (3 pattern words)

3.5 Address

The Address is provided in word I as bits 4 to 19 (16 bits). This indicates which channel on which module has provided this event fragment.
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF0000000	>>	28	=	0x8	=	8

An example of unpacking the Address from the header word:

Word I		Mask		Shift		Hex result		Decimal result
0x82228280	&	0x000FFFF0	>>	4	=	0x0505	=	1285

The various properties with the Address word are as follows:

Property	Size	Values	Description
M	4 bits	0 to 15	Input channel to Master or ID number of Slave
S	4 bits	0 to 15	Input channel to Slave or ID number of Digitizer
C	8 bits	0 to 255	Input channel to Digitizer

These properties can be extracted from this 16 bit word in the following way:

Address		Mask		Shift		Hex result		Decimal result
0x1405	&	0x0000F000	>>	12	=	0x01	=	1 (M)
0x1405	&	0x00000F00	>>	8	=	0x04	=	4 (S)
0x1405	&	0x000000FF	>>	0	=	0x05	=	5 (C)

3.6 Detector Type

The Detector Type is provided in word I as bits 0 to 3 (4 bits). This indicates which detector type has generated this event fragment and is used primarily in master filter decisions.
Example of identifying a Header word type:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A928	&	0xF0000000	>>	28	=	0x8	=	8

The Detector Type value can be extracted as:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A900	&	0x0000000F	>>	0	=	0x00	=	00 (HPGe Low Gain)
0x8074A902	&	0x0000000F	>>	0	=	0x02	=	02 (SCEPTAR)
0x8074A905	&	0x0000000F	>>	0	=	0x05	=	05 (Si(Li))

Here is a full list of GRIFFIN Detector Types:

ID	**Short Name**	**Full Name**	**Description**
00	GRGa	Ge	16 HPGe Clovers Low Gain
01	GRGb	Ge	16 HPGe Clovers High Gain
02	SEP	Beta	20 plastic scintillators (SCEPTAR)
03	DAN	DANTE	8 LaBr3 scintillators Energy signals
04	DAT	DANTE	8 LaBr3 scintillators TAC signals
05	PAC	PACES	5 SiLi for ICE
06	DSC	DESCANT	70 Scintillators for neutrons
07	GRS	Suppressors	16 sets of BGO for clovers
08	DAS	Suppressors	8 sets of BGO for DANTE
09	SET	Beta	Zero-Degree scintillator (ZDS)
10	DSC	DESCANT (Gamma)	Gamma identified by pulse shape
11	DSC	DESCANT (Neutron)	Neutron identified by pulse shape
12			Undefined
13			Undefined
14			Undefined
15			Undefined

3.7 Master Filter ID

The Master Filter ID is provided in word III as 31 bits. The Master Filter ID is an event counter which is reset at the start of a run and increments by 1 each time a new event master filter condition is passed in the master filter algorithm in the

Master GRIF-C. The maximum value is 2147483649 above which the counter will wrap around.
Example of extracting the Master Filter ID:

Word II		Mask		Shift		Hex result		Decimal result
0x56007C84	&	0x7FFFFFFF	>>	0	=	0x56007C84	=	1442872452

3.8 Master Filter Pattern

There are up to eight (more in the future) Filter Patterns which can be set and tested for every fragment. The Identifier(s) (1-8) of the Filter Pattern which was passed is provided in word II as bits 16 to 30 (14 bits). If multiple Filter conditions were passed by this fragment then there will be multiple Filter Patterns indicated in this word.
Example of extracting the Master Filter Pattern(s):

Word III		Mask		Shift		Hex result		Decimal result
0x00010F81	&	0x7FFF0000	>>	16	=	0x0001	=	1 (Only Filter 1 passed)
0x00050F81	&	0x7FFF0000	>>	16	=	0x0005	=	5 (Filters 1 and 3 passed)
0x00850F81	&	0x7FFF0000	>>	16	=	0x0085	=	133 (Filters 1, 3 and 8 passed)

3.9 PPG Pattern

The PPG Pattern is a recording of the Programable Pattern Generator status at the time that the threshold crossing occurred, and is provided in word II as bits 0 to 15 (16 bits). The pattern represents, for example, if the beam were being delivered or if it was blocked for decay counting etc.
Example of extracting the PPG Pattern:

Word III		Mask		Shift		Hex result		Decimal result
0x0F810F81	&	0x0000FFFF	>>	0	=	0x00000F81	=	3969

3.10 Channel Trigger ID

The Channel Trigger ID is provided in word IV as the 28 lower bits. The Channel Trigger ID is an event counter which is reset at the start of a run and increments by 1 each time a new threshold crossing occurs for this channel on the GRIF-16 or GRIF-4G. The word type is identified in the 4 upper bits as type 9.
Example of identifying the Channel Trigger ID word type:

Word IV		Mask		Shift		Hex result		Decimal result
0x900461FF	&	0xF0000000	>>	28	=	0x9	=	9

The Channel Trigger ID number can wrap around if it becomes too large. The user should monitor the channel trigger id numbers which should increase throughout the run. The maximum value of channel trigger id is 268435455. If the number changes from very large to small then the counter has wrapped around, this should be tested at every event.
Example of unpacking the Channel Trigger ID:

Word IV		Mask		Shift		Hex result		Decimal result
0x900461FF	&	0x0FFFFFFF	>>	0	=	0x000461FF	=	287231

3.11 Timestamp

The timestamp is stored in two separate words as the lower 28 bits and 14 bits of the two words, word V and word VI respectively. Word V is the lower bits of the Timestamp and Word VI is the upper bits of the Timestamp. The type is identified in the 4 upper bits of both words as type 10 for the lower bits and 11 for the upper bits.
Example of identifying a Timestamp word type:

Word V/VI		Mask		Shift		Hex result		Decimal result
0xA074A998	&	0xF0000000	>>	28	=	0xA	=	10 (lower bits)
0xB074A998	&	0xF0000000	>>	28	=	0xB	=	11 (upper bits)

An example of unpacking the Timestamp value:

Word V/VI		Mask		Shift		Hex result		Decimal result
0xA074A998	&	0x0FFFFFFF	>>	0	=	0x0074A998	=	7645592
0xB0000005	&	0x00003FFF	>>	0	=	0x00000005	=	5

The upper and lower bits of the Timestamp can then be combined to make the full 42-bit timestamp. This is done in two steps by shifting the upper bits and then adding the lower bits using a bitwise or (||):

Build Timestamp				Hex result		Decimal result
0x00000005	<<	28	=	0x05000000		
0x05000000	||	0x0074A998	=	0x0574A998	=	91531672

The timestamp is in units of 10 nanoseconds and has a maximum value of 4,398,046,511,103. The timestamp will therefore reset either after 509 days or when the GRIF-C Master module is power cycled.

The 28 lower bits are also stored in the event trailer.

3.12 Deadtime

There are two methods of determining the event-by-event deadtime of the system for precision measurements; firstly to operate at a fixed deadtime for every event, or

secondly to measure and record the deadtime of each event. Both methods will be trialled and evaluated as part of the GRIFFIN DAQ development.

The deadtime for each event is provided as bits 14-27 (14 bits) of word VII. The type is identified in the 4 upper bits as type 11.
Example of identifying a Deadtime word type:

Word VII		Mask		Shift		Hex result		Decimal result
0xB074A998	&	0xF0000000	>>	28	=	0xB	=	11

An example of unpacking the Deadtime value:

Word VII		Mask		Shift		Hex result		Decimal result
0xC074A998	&	0x0FFFFC000	>>	14	=	0x0074A998	=	7645592

The deadtime is in units of 10 ns and the maximum value is 16,383 representing 162.383 microseconds of deadtime for this event readout.

3.13 Network Packet ID

The Network Packet ID is currently provided in word VIa as the 28 lower bits. The Network Packet ID is an event counter which is reset at the start of a run and increments by 1 each time a new event is dispatched by the GRIF-C master. The word type is identified in the 4 upper bits as type 9.
Example of identifying the Network Packet ID word type:

Word VIa		Mask		Shift		Hex result		Decimal result
0xD00461FF	&	0xF0000000	>>	28	=	0xD	=	13

The Network Packet ID number can wrap around if it becomes too large. The user should monitor the Network Packet id numbers which should increase throughout the run. The maximum value of Network Packet ID is 268435455. If the number changes from very large to small then the counter has wrapped around, this should be tested at every event.
Example of unpacking the Network Packet ID:

Word VIa		Mask		Shift		Hex result		Decimal result
0xD00461FF	&	0x0FFFFFFF	>>	0	=	0x000461FF	=	287231

3.14 Waveform Data

The Waveform Data is provided in multiple words, the number of which depends on the waveform length. These words are inserted into the event between words VI and VII. Waveform data words can be identified by their type which is in the 4 upper bits of each words as type C.

Word		Mask		Shift		Hex result		Decimal result
0xC57715CC	&	0xF0000000	>>	28	=	0xC	=	12

The Waveform Data values are in units of ADC bins.

3.15 K Values

The 10 bits of the K value are provided in words VII and VIII as bits 26-30.
Example of extracting the K value:

Word VII/VIII		Mask		Shift		Hex result		Decimal result
0x2074A998	&	0x7C000000	>>	26	=	0x08	=	8
0x5874A998	&	0x7C000000	>>	26	=	0x16	=	22

The upper and lower bits of the K value can then be combined to make the full K value. This is done in two steps by shifting the upper bits and adding the lower bits:

Build K Value				Hex result		Decimal result
0x08	<<	5	=	0x100		
0x100	||	0x16	=	0x116	=	278

3.16 Pulse Height Signal

The pulse height signal is provided in word VII as the 26 lower bits, which represent Pulse height 1. The maximum value is 67108863 and is in units of ADC channels.

Two examples of unpacking the Pulse Height value:

Word VII		Mask		Shift		Hex result		Decimal result
0x680001FF	&	0x03FFFFFF	>>	0	=	0x000001FF	=	511
0x680005B5	&	0x03FFFFFF	>>	0	=	0x000005B5	=	1461

There is no handling of pile-up in the GRIF-4G module. Pile-up events will cause occasional bad data.

3.17 CFD Timing Signal

Note that Word VIII and IX are only read out for the DESCANT detector type.

The timing from a CFD algorithm is provided in two parts word VIII as the 21 lower bits. There is both a CFD time itself (bits 0-20) and the "remainder" (bits 21-24) which is generated because of the difference between the 100 MHz timestamp and the 125 MHz clock that the FPGA is running on. The CFD time is in units of 1/256 ns.

A time in units of 1 ns can be constructed as $[(\text{Timestamp} \times 10) + \text{Remainder}]$.

A time in steps of 1/256 ns ($\sim$ 3.9 ps) can be constructed as [(Timestamp $\times$ 10) + Remainder + $\frac{\text{CFD}}{256}$].

3.18 Short Charge Integration

Note that Word VIII and IX are only read out for the DESCANT detector type.

The 19 bits of the Short Charge Integration are provided in word IX as bits 0-18. This short integration of the charge signal is used in combination with the Long Charge Integration and the CFD information to provide neutron-gamma discrimination. The word type is identified in the 4 upper bits as type 0xf.
Example of identifying the Short CHarge Integration word type:

Word IX		Mask		Shift		Hex result		Decimal result
0xF00461FF	&	0xF0000000	>>	28	=	0xF	=	15

Two examples of unpacking the Short Charge Integration value:

Word IX		Mask		Shift		Hex result		Decimal result
0xF00461FF	&	0x001FFFFF	>>	0	=	0x000001FF	=	511
0xF00465B5	&	0x001FFFFF	>>	0	=	0x000005B5	=	1461

3.19 Long Charge Integration

Note that Word VIII and IX are only read out for the DESCANT detector type.

The 19 bits of the Long Charge Integration are provided in words IX as bits 19-27 and X as bits 22-31.
Example of extracting the Long Charge Integration:

Word IX/X		Mask		Shift		Hex result		Decimal result
0xF044A5B5	&	0x0FF80000	>>	19	=	0x08	=	8
0x02C4A998	&	0x7FE00000	>>	21	=	0x16	=	22

The upper and lower bits of the Long Charge Integration can then be combined to make the full Long Charge Integration. This is done in two steps by shifting the upper bits and adding the lower bits:

Build Long Integration				Hex result		Decimal result
0x0016	<<	9	=	0x2000		
0x2000	||	0x08	=	0x2008	=	8200

3.20 Zero-Crossing Time

Note that Word VIII and IX are only read out for the DESCANT detector type.

The Zero-Crossing Time is provided in word X as the 21 lower bits. This is the result of a CFD algorithm processing the signal after it has been integrated.

Two examples of unpacking the Zero-Crossing Time value:

Word X		Mask		Shift		Hex result		Decimal result
0x78000052	&	0x001FFFFF	>>	0	=	0x00000052	=	82
0x780000F6	&	0x001FFFFF	>>	0	=	0x000000F6	=	246

3.21 Event Trailer

The Event Trailer is provided in word XI and it indicates the end of the event fragment. The type is identified in the 4 upper bits as type 14.
Example of identifying the Event Trailer word type:

Word XI		Mask		Shift		Hex result		Decimal result
0xEEEEEEEE	&	0xF0000000	>>	28	=	0xE	=	14

The 14 lower bits of the timestamp is repeated in the event trailer. This can be used as an additional check to ensure the event is not corrupted. An example of unpacking the lower bits of the Timestamp value from the Event Trailer word:

Word XI		Mask		Shift		Hex result		Decimal result
0xE074A998	&	0x0003FFFF	>>	0	=	0x0000A998	=	43416

Chapter 4

GRIF-C

Not yet fully defined.

Additional information will be produced from the GRIF-C such as statistics and information on any events which have been dropped and did not make it to the disk file.

Chapter 5

5.1 List of Packet Type Identifiers

The first 4 bits of some 32-bit words contain a packet type identifier. These can be used to ensure correct identification of data words and appropriate use of the words.

An example of identifying a Header word type, the same method is used to get all packet type identifiers:

Word I		Mask		Shift		Hex result		Decimal result
0x8074A998	&	0xF000000	>>	28	=	0x8	=	8

A full list of packet types is listed here:

Packet Type (Hexadecimal)	Packet Type (Decimal)	Name
0x8	8	Header word (Start of event)
0x9	9	Channel Trigger ID word
0xA	10	Timestamp Lower bits word
0xB	11	Deadtime and Timestamp Upper bits word
0xC	12	Waveform sample
0xD	13	Network Packet ID
0xE	14	Trailer word (End of event)
0xF	15	Lower bits of Long Charge Integration and Short Charge Integration

5.2 List of Bit Masks and Bit Shifts

A full list of bit masks is given here. See individual sections for the meanings:

Mask	Shift	Name
0x0C000000	26	Pile-Up Type
0x03800000	23	Data Type
0x00700000	20	Number of Filter Patterns
0x000FFFF0	4	Address
0x0000000F	0	Detector Type
0x7FFF0000	0	Master Filter Patterns Passed
0x0000FFFF	0	PPG Pattern
0x7FFFFFFF	0	Master Filter ID
0x0FFFFFFF	0	Channel Trigger ID
0x0FFFFFFF	0	Network Packet ID
0x0FFFFFFF	0	Timestamp Lower Bits
0x00003FFF	0	Timestamp Upper Bits
0x0FFFC000	14	Deadtime
0x7C000000	26	K Value Lower Bits
0x7C000000	26	K Value Upper Bits
0x03FFFFFF	0	Pulse Height Signal
0x03FFFFFF	0	CFD Timing Signal
0x001FFFFF	0	Zero-Crossing Time
0x001FFFFF	0	Short Charge Integration
0x0FF80000	19	Long Charge Integration Upper Bits
0x7FE00000	21	Long Charge Integration Lower Bits